

T M V S G Pavan

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EDUCATION

Birla Institute of Technology and Science, Pilani (Goa Campus) | 2024 – 2026

M.E. Computer Science (CGPA: 9.55) · Goa

Sri Venkateswara University College of Engineering | 2020 – 2024

B.Tech Computer Science & Engineering (CGPA: 8.6) · Tirupati

EXPERIENCE

Netskope | Jan 2026 – Present

AI Intern – Intrusion Prevention System (IPS) Team · Bangalore

- Designed a multi-agent AI triage pipeline using LangChain tool-calling agents – orchestrator agent delegates to specialized tool agents (VirusTotal reputation, domain WHOIS age, User-Agent anomaly scoring, process behavior profiling) to autonomously classify tens of thousands of weekly IPS security alerts, achieving 80% reduction in manual analyst workload.
- Built a production ML pipeline ingesting 4.25M events from PostgreSQL: multi-stage label conflict resolution, stratified downsampling, and 72 engineered features via TF-IDF + Truncated SVD (Latent Semantic Analysis) embeddings across URL, User-Agent, process, and hostname fields.
- Trained LightGBM, XGBoost, Random Forest, SVM, LSTM, Bi-LSTM, GRU, FFNN with Optuna; tracked experiments via MLflow; deployed as Docker-containerized FastAPI microservice with a drift-monitoring agent for automated retraining on evolving threat patterns.

PROJECTS

Phishing Detection with AI Agents & Explainable DL | PyTorch, RoBERTa, LangChain, FastAPI, Docker | Aug 2024 – Dec 2024

- Built an agentic phishing detection pipeline: LangChain orchestrator delegates URL feature extraction, transformer embedding generation (BERT-CLS, RoBERTa, SBERT, USE, ALBERT), and model selection to specialized tool agents – deployed as a FastAPI microservice in Docker achieving 99.9% accuracy with RoBERTa-LSTM.
- Trained FFNN, RNN, LSTM, Bi-LSTM, GRU on standalone vs. concatenated embeddings; built an explainability agent that invokes Captum Integrated Gradients and SHAP/LIME to generate human-readable threat attribution reports for SOC analysts.

BITS Social – Production Social Media Backend | TypeScript, Express.js, PostgreSQL, Redis, Kafka, Docker | Apr 2025 – Present

- Architected a scalable Twitter-clone backend: Express.js 5, Prisma ORM, PostgreSQL/CockroachDB (3-node cluster), Redis sorted-set timelines, Kafka fan-out-on-write – containerized via Docker Compose (6 services), deployed on Vercel; integrated AI content moderation agent for automated post screening.
- Built JWT + Google OAuth 2.0, Redis sliding-window rate limiter, S3 media storage (MinIO), Zod validation, structured logging – 10+ RESTful modules for feeds, posts, follows, polls, and real-time notifications.

3D NoC Mapping & TSV Optimization (M.E. Thesis) | Python, Gurobi ILP, Noxim Simulator | Feb 2025 – Nov 2025

- Multi-objective optimization of IP core mapping + TSV placement in 3D Network-on-Chip; minimized communication cost and thermal hotspots; validated on Noxim-generated production-grade traffic traces.
- Implemented Tabu Search (with agent-guided parameter tuning) and ILP (Gurobi); Tabu achieved 95–98% of optimal quality with 10x runtime reduction; outputs integrated into downstream EDA toolchain.

TECHNICAL SKILLS

Languages: Python, C++, C, TypeScript, SQL

ML/AI: PyTorch, scikit-learn, LightGBM, XGBoost, SVM, LSTM, Bi-LSTM, GRU, CNN, NLP, Transformers (BERT, RoBERTa), LangChain, AI Agents, RAG, SHAP, Captum, Optuna, MLflow, MLOps

Backend & DevOps: Node.js, Express.js, FastAPI, REST APIs, Prisma, Kafka, Redis, Docker, Vercel, AWS S3, CI/CD

Databases: PostgreSQL, CockroachDB, MySQL · Tools: Git, Linux, DBeaver

PUBLICATIONS

- PhishBuster | EAI Broadnets 2024 | Published: Jan 2026 | DOI: 10.1007/978-3-032-14350-1_4